

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS student (NC State, May 2026) with production full-stack experience in TypeScript, React, and Node.js -- the exact stack Replit runs on. At Isomer AI, shipped 10+ features across 7 production deployments including multi-model AI agent tooling (Claude, GPT-4o, Gemini via MCP) and real-time event pipelines -- 394 commits to a live production codebase. Built AlgoPulse, an AI orchestration platform handling 2,000 async inference requests/second using Temporal.io workflows and Kafka. Designed Glance AI, a compliance-ready RAG system for fintech/healthcare. Comfortable building and deploying end-to-end: from AI agent backend to polished React UI to cloud infrastructure on GCP and AWS.

CORE COMPETENCIES

TypeScript & React

Node.js / Express APIs

AI Agent Integration (MCP / LLMs)

Full-Stack Product Engineering

Cloud Infrastructure (GCP / AWS)

Real-Time Systems (Pusher / WebSockets)

Distributed Workflows (Temporal / Kafka)

Python & AI Pipelines

CI/CD & Docker

EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern - Remote

- Shipped **10+ production features** in a **TypeScript/React/Node.js/PostgreSQL** stack across 7 deployments -- building real-time event pipelines (Pusher), document processing workflows, RBAC flows, and multi-model AI agent tooling end-to-end
- Integrated **MCP tool-calling** across Claude, GPT-4o, and Gemini; wired Temporal.io for durable AI workflow orchestration; used Auth0, Redis, Docker, S3, and Sentry across the full stack -- **394 commits** to production
- Owned features from spec to deployment: wrote clean TypeScript, participated in code review, and maintained zero-downtime across 7 releases in a fast-moving startup environment

NC State University Apr 2025 - Present

Graduate Research Assistant

- Built a **React + JBrowse 2** genomic data platform deployed on **Google Cloud Run** serving 100+ researchers across 500GB of datasets; applied ML hyperparameter tuning improving model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven automation workflows in Python; automated deployments on AWS & GCP with Grafana monitoring dashboards; improved health claim accuracy by 25%

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Built Kafka streaming + Spark batch pipelines for real-time support ticket classification; reduced latency by 20%

PROJECTS

AlgoPulse - AI Orchestration Platform Independent · 2025

Built a **full-stack AI platform** in TypeScript/React/Node.js for agentic analysis. Identified a polling bottleneck under load and re-architected to event-driven streaming via Supabase Edge Functions, **Temporal.io** durable workflows, and Kafka state sync. Scaled to **2,000 async AI inference requests/second** with zero state loss under fault conditions.

TypeScript · React · Node.js · Temporal.io · Kafka · Supabase Edge Functions · PostgreSQL

Glance AI - Privacy-First RAG System Independent · 2025

Compliance-ready RAG system for fintech and healthcare with **per-tenant data isolation** at the vector retrieval layer -- 0 cross-tenant context leakage by design. Deterministic retrieval architecture eliminates hallucinations for regulated environments.

Python · LLMs · Vector Databases · RAG

CryptoKnight - Trustless AI Trading Agent HackNC State · 2025

Autonomous Polymarket trading agent using **ERC-4337 Account Abstraction and Session Keys** -- 100% autonomous, 0 private keys shared. Built **Next.js 16 + viem + wagmi** frontend with Python agent backend. Session Keys enforce on-chain trade limits, allowance caps, and expiration -- funds stay in self-custody vault.

Next.js 16 · viem · wagmi · Python · ERC-4337 · Alchemy Account Kit

Sanjeevani - Health App Top 1% of 1,100+ teams

Full-stack **React** health application with ML-powered predictive insights achieving 87% accuracy on real-time sensor data. Shortlisted top 12 of 1,100+ teams for UI quality and product innovation.

React · Node.js · Python · ML classification models

EDUCATION

North Carolina State University - MS Computer Science	Aug 2024 - May 2026
Courses: Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms	
Mumbai University - BTech Computer Science	Aug 2020 - May 2024
CGPA: 9.32/10 · Courses: DS & Algorithms, Machine Learning, Software Engineering, Big Data Infrastructure	

SKILLS

- Frontend:** React (v16+), TypeScript, JavaScript (ES6+), Next.js, Tailwind CSS, Vite, HTML/CSS
- Backend:** Node.js, Express, Python, PostgreSQL, Redis, Prisma, REST APIs
- Infrastructure:** AWS, GCP, Docker, CI/CD, Kafka, Temporal.io, Git, Google Cloud Run
- AI/ML:** OpenAI API, Anthropic Claude, LangChain, MCP, RAG, PyTorch, TensorFlow, Hugging Face
- Other:** Java, C, SQL, Grafana, Sentry, Auth0, S3, Pusher, viem, wagmi